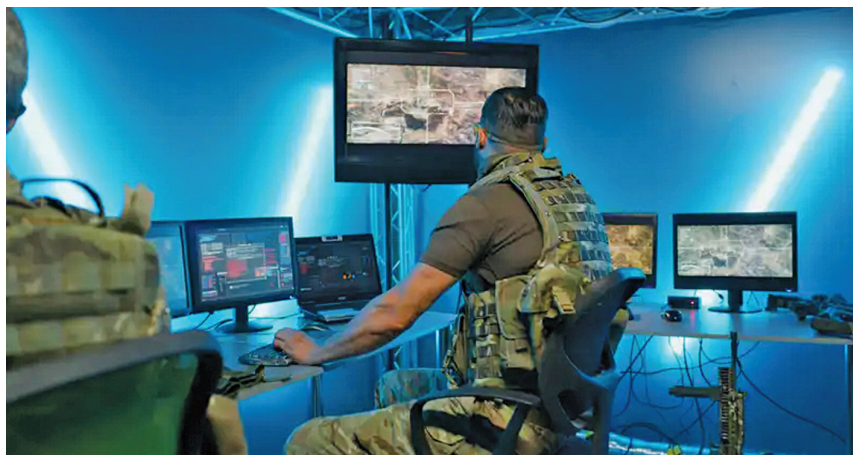
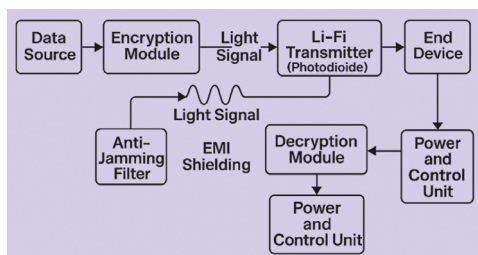




Li-Fi provides point-to-point communication links to the defence forces that are hard to detect (Source: <https://www.oledcomm.net>)



Secure Li-Fi optical communication system developed by DRDO

with IIT Delhi and IIT Madras, has developed an indoor Li-Fi navigation system that delivers precise position tracking for military forces in such environments. These include areas where communication signals may be hindered, as well as locations such as trenches and temporary shelters where intrusion risks are high.

The system includes a network of ceiling-mounted LED transmitters broadcasting encoded location signals as optical beacons confined to the indoor environment. Soldiers carry photodiode receivers integrated into handheld devices or helmet-mounted units. The receivers detect the LED transmissions, convert the light pulses into electrical signals, decode the embedded codes, and translate them into precise positional coordinates.

The positional coordinates are then fused with readings from inertial sensors and magnetometers to maintain stable navigation when

signals are partially obstructed or when interference levels are high. The decoded coordinates are displayed on units carried by soldiers to facilitate movement through corridors, basements, and shelters. Field trials are now in progress.

ISRO: Integrating Li-Fi into satellites for military operations. ISRO is integrating Li-Fi into satellites to establish high-speed wireless links in space capable of delivering up to 2Gbps over distances of 1km and handling large volumes of data

during orbital operations. Li-Fi also reduces satellite weight by replacing heavy cables and connectors with lighter optical links.

The upcoming TDS-01 satellite mission will serve as a testbed to validate these optical links in orbit, enabling a future 'space mesh' network designed to keep strategic assets seamlessly connected during critical military operations.

Li-Fi experimental testbed for the defence forces. ERNET and IIT Madras have jointly developed a Li-Fi experimental testbed that serves as a research and evaluation platform operating under line-of-sight and non-line-of-sight conditions for defence applications.

The testbed is designed to continuously measure received optical power, analyse outage behaviour, and assess link stability and performance. It can conduct evaluations in



A military drone inspecting a tunnel using Li-Fi in a GPS-denied environment (Source: <https://militaryembedded.com>)



Vehicular communication systems for defence fleets (Source: <https://lifimax.ch>)